



Vehicle V_i
OBU



TA

Key generation (BN254 curve)

1

Generate long-term key pair:

$$(sk_i, pk_i) \leftarrow \text{KeyGen}(\mathbb{G}_1)$$

2

VIN_i, pk_i

secure out-of-band channel

Privacy property

Blind signature ensures σ_i is
unlinkable to VIN_i for all parties
except TA.

Verification & credential issuance

Verify V_i 's legitimacy via VIN_i

Compute a blind signature:

$$\sigma_i \leftarrow \text{Sign}_{sk_{TA}}(H(pk_i || \text{nonce}))$$

(unlinkable to the real identity)

3

σ_i, pk_{TA}

anonymous credential

Secure storage (tamper-proof module)

4

Store: sk_i, σ_i, pk_{TA}

pk_{TA} used later to verify AIS certificates.